

Pollution Control

To control pollution, the below said three techniques are being used,

a) Reduce pollution before it occurs
b) Reduce the pollutant during their formation stages
c) Reduce pollution once it is formed.

(1) Reduce pollution before it occurs,
- Low compression ratio
- Changing the combustion chamber geometry
- Changing the piston design
- Lean mixture operation
- Maintaining piston and piston ring function

(2) Reduce the pollutant during their formation stages
- Fuel modifications
- Engine modifications
- Modifying the operating parameters for complete combustion

(3) Reduce pollution once it is formed,
- Using burner
- More air supply to inlet manifold

(4) Control of carbon monoxide,
- Lean mixture operation
- After burner
- Catalytic converter

(5) Control of oxides of nitrogen,
- Re inducting the exhaust gas into inlet manifold
- Mixing of non-fired things with fired things

Smoke Control

- To control the smoke we have to maintain the vehicle in good condition and barium salt to the fuel

Various Methods of Reducing Emissions

- Re setting of valve timing
- Inspecting cooling and fuel supply system
- Changing combustion chamber design
- Re inducting the exhaust gas into inlet manifold
- lean mixture operation

Student Activity

1. Students should visit the nearby workshops to study intake and exhuast system of a motorcycle and a car and should have a hands-on experience on engine dismantling and assembling inlet manifold.
2. Students should visit the nearby workshops to dismandle the cylinder head and sketch the components.

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